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What is a Solenoid valve and how does it work?

8-10 minutes

A Solenoid valve is used wherever fluid flow has to be controlled automatically. They are being used to an increasing degree in the most varied types of plants and equipment. The variety of different designs which are available enables a valve to be selected to specifically suit the application in question.

How is a solenoid operated valve being made?

There are 5 main parameters to consider when selecting a valve:

- Cv
- media compatibility
- pressure
- temperature
- process fitting

Solenoid valve is a control units which, when electrically energized or de-energized, either shut off or allow fluid flow. The actuator takes the form of an electromagnet. When energized, a magnetic field builds up which pulls a plunger or pivoted armature against the

action of a spring. When de-energized, the plunger or pivoted armature is returned to its original position by the spring action.

How does a solenoid valve operate?

To the mode of actuation, a distinction is made between direct-valves, internally piloted valves, and externally piloted valves. A further distinguishing feature is the number of port connections or the number of flow paths ("ways").

Direct-acting solenoid valve

With a direct-acting solenoid valve, the seat seal is attached to the solenoid core. In the de-energized condition, a seat orifice is closed, which opens when the valve is energized.

Direct-acting 2-way solenoid valve

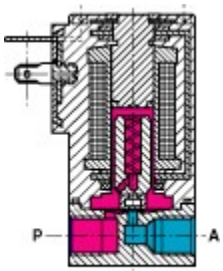


figure 1

Two-way solenoid operated valves are shut-off valves with one inlet port and one outlet port (Fig. 1). In the de-energized condition, the core spring, assisted by the fluid pressure, holds the valve seal on the valve seat to shut off the flow. When energized, the core and seal are pulled into the solenoid coil and the valve opens. The electro-magnetic force is greater than the combined spring force and the static and dynamic pressure forces of the medium.

Direct-acting 3-way solenoid valve operation

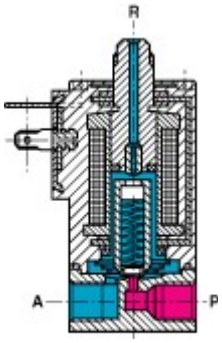


figure 2

Three-way solenoid operated valves have three port connections and two valve seats. One valve seal always remains open and the other closed in the de-energized mode. When the coil is energized, the mode reverses. The 3-way solenoid valve shown in Fig. 2 is designed with a plunger type core. Various valve operations can be obtained according to how the fluid medium is connected to the working ports in Fig. 2. The fluid pressure builds up under the valve seat. With the solenoid coil de-energized, a conical spring holds the lower core seal tightly against the valve seat and shuts off the fluid flow. Port A is exhausted through R. When the coil is energized the core is pulled in, the valve seat at Port R is sealed off by the spring-loaded upper core seal. The fluid medium now flows from P to A.

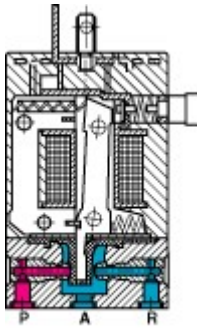


figure 3

Unlike the versions with plunger-type cores, pivoted-armature valves have all port connections in the valve body. An isolating diaphragm ensures that the fluid medium does not come into contact with the solenoid coil

chamber. Pivoted-armature valves can be used to obtain any 3-way valve operation. The basic design principle is shown in Fig. 3. Pivoted-armature valves are provided with manual override as a standard feature.

Internally piloted solenoid valve

With direct-acting valves, the static pressure forces increase with increasing orifice diameter which means that the magnetic forces, required to overcome the pressure forces, become correspondingly larger. Internally piloted solenoid valves are therefore employed for switching higher pressures in conjunction with larger orifice sizes; in this case, the differential fluid pressure performs the main work in opening and closing the valve.

Internally piloted 2-way solenoid valve

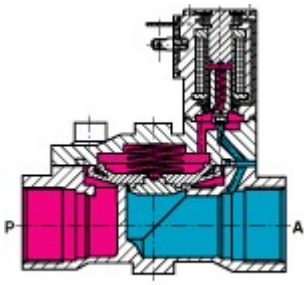


figure 4

Internally piloted solenoid valves are fitted with either a 2- or 3-way solenoid valve. A diaphragm or a piston provides the seal for the main valve seat. The operation of such a valve is indicated in Fig. 4. When the pilot valve is closed, the fluid pressure builds up on both sides of the diaphragm via a bleed orifice. As long as there is a pressure differential between the inlet and outlet ports, a shut-off force is available by virtue of the larger effective area on the top of the diaphragm. When the pilot valve is opened, the pressure is relieved from the upper side of the

diaphragm. The greater effective net pressure force from below now raises the diaphragm and opens the valve. In general, internally piloted valves require a minimum pressure differential to ensure satisfactory opening and closing. Omega also offers internally piloted valves, designed with a coupled core and diaphragm that operate at zero pressure differential (Fig. 5).

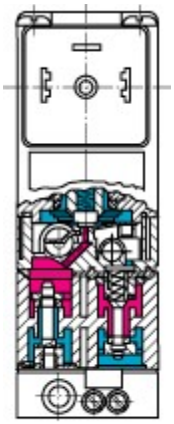


figure 6

Internally piloted multi-way solenoid valve

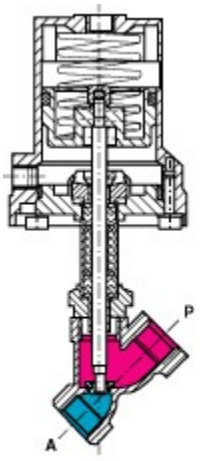


figure 7

Internally piloted 4-way solenoid valves are used mainly in hydraulic and pneumatic applications to actuate double-acting cylinders. These valves have four port connections: a pressure inlet P, two cylinder port connections A and B, and one exhaust port connection R. An internally piloted 4/2-way poppet valve is shown

in Fig. 6. When de-energized, the pilot valve opens at the connection from the pressure inlet to the pilot channel. Both poppets in the main valve are now pressurized and switch over. Now port connection P is connected to A, and B can exhaust via a second restrictor through R.

Externally piloted solenoid valve

With these types an independent pilot medium is used to actuate the valve. Fig. 7 shows a piston-operated angle-seat valve with closure spring. In the unpressurized condition, the valve seat is closed. A 3-way solenoid valve, which can be mounted on the actuator, controls the independent pilot medium. When the solenoid operated valve is energized, the piston is raised against the action of the spring and the valve opens. A normally-open valve version can be obtained if the spring is placed on the opposite side of the actuator piston. In these cases, the independent pilot medium is connected to the top of the actuator. Double-acting versions controlled by 4/2-way valves do not contain any spring.

[How to select a Solenoid valve for process applications]

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